TASK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS

PROJECT DETAILS

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SCENARIO

The Artisan Crumb Bakery & Pastry Management System stores customer, bakery item, order, and payment information in separate tables. The management team wants to combine these related tables to understand customer purchases, order details, and payment status.

Implement INNER JOIN

Concept

INNER JOIN returns only matching records from both tables.

Tasks

1. Display customer details along with their orders.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  INNER JOIN Orders O
  8    ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- --------- ------------
C004                 Bob White
ORD-505              24-MAR-26         1250

C002                 Jane Smith
ORD-506              25-MAR-26          720

C003                 Alice Brown
ORD-507              26-MAR-26         1450


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- --------- ------------
C001                 John Doe
ORD-508              27-MAR-26          890
```

2. Display order details with payment information.

```
SQL> SELECT O.Order_ID,
  2         O.Total_Amount,
  3         O.Order_Date,
  4         O.Customer_ID,
  5         P.Payment_Date,
  6         P.Payment_Mode,
  7         P.Payment_Status
  8  FROM Orders O
  9  INNER JOIN Payment P
 10    ON O.Order_ID = P.Order_ID;

ORDER_ID             TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
-------------------- ------------ --------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS
--------------------------------------------------
ORD-507                      1450 26-MAR-26 C003                 26-MAR-26
Debit Card
Failed

ORD-507                      1450 26-MAR-26 C003                 26-MAR-26
Credit Card
Completed

ORDER_ID             TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
-------------------- ------------ --------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS
--------------------------------------------------

ORD-508                       890 27-MAR-26 C001                 27-MAR-26
Debit Card
Completed

ORD-505                      1250 24-MAR-26 C004                 24-MAR-26
UPI

ORDER_ID             TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
-------------------- ------------ --------- -------------------- ---------
PAYMENT_MODE
--------------------------------------------------
PAYMENT_STATUS
--------------------------------------------------
Completed

ORD-506                       720 25-MAR-26 C002                 25-MAR-26
COD
Completed
```

3. Retrieve bakery items purchased by customers.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         O.Order_ID,
  4         P.Product_ID,
  5         P.Title AS Bakery_Item_Name,
  6         OD.Quantity,
  7         OD.Price,
  8         (OD.Quantity * OD.Price) AS Total_Amount
  9  FROM Customer C
 10  JOIN Orders O
 11    ON C.Customer_ID = O.Customer_ID
 12  JOIN Order_Details OD
 13    ON O.Order_ID = OD.Order_ID
 14  JOIN Product P
 15    ON OD.Product_ID = P.Product_ID
 16  ORDER BY C.Customer_ID, O.Order_Date;

CUSTOMER_ID          CUSTOMER_NAME                          ORDER_ID             PRODUCT_ID
-------------------- -------------------------------------- -------------------- ----------
BAKERY_ITEM_NAME
--------------------------------------------------------------------------------------------
  QUANTITY    PRICE TOTAL_AMOUNT
---------- -------- ------------
C002                 Jane Smith                             ORD-506              P001
Chocolate Fudge Cake
         1   750.00          750

C004                 Bob White                              ORD-505              P005
Black Forest Cake
         2   600.00         1200
```

Implement LEFT JOIN

Tasks

1. Display all customers including customers who have not placed orders.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount,
  6         O.Order_Status
  7  FROM Customer C
  8  LEFT JOIN Orders O
  9    ON C.Customer_ID = O.Customer_ID
 10  ORDER BY C.Customer_ID;

CUSTOMER_ID          CUSTOMER_NAME                          ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------------------------- -------------------- --------- ------------
ORDER_STATUS
------------------------------------------------
C001                 John Doe                               ORD-508              27-MAR-26          890
Pending

C002                 Jane Smith                             ORD-506              25-MAR-26          720
Completed

C003                 Alice Brown                            ORD-507              26-MAR-26         1450
Baking

C004                 Bob White                              ORD-505              24-MAR-26         1250
Completed

C005                 Charlie Green                          ORD-509              28-MAR-26          950
Completed
```

2. Find customers without any purchases.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         C.Email,
  4         C.Phone
  5  FROM Customer C
  6  LEFT JOIN Orders O
  7    ON C.Customer_ID = O.Customer_ID
  8  WHERE O.Order_ID IS NULL
  9  ORDER BY C.Customer_ID;

CUSTOMER_ID         CUSTOMER_NAME                          EMAIL                         PHONE
------------------- -------------------------------------- ----------------------------- ----------
C006                David Miller                           david.miller@email.com         9876543210
```

3. Display all bakery items including bakery items with no sales.

```
SQL> SELECT P.Product_ID,
  2         P.Title,
  3         P.Price
  4  FROM Product P
  5  LEFT JOIN Order_Details OD
  6    ON P.Product_ID = OD.Product_ID
  7  WHERE OD.Product_ID IS NULL
  8  ORDER BY P.Product_ID;

PRODUCT_ID TITLE                          PRICE
---------- ------------------------- --------
P002       Sourdough Bread             350.00
P003       Red Velvet Pastry           450.00
P004       Croissants                  200.00
P006       Special Wedding Cake       1200.00
```

Implement RIGHT JOIN

Concept

RIGHT JOIN returns all records from the right table and matching records from the left table.

Tasks

1. Display all orders with customer information.

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         O.Total_Amount,
  4         O.Order_Status,
  5         C.Customer_ID,
  6         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  7         C.Email
  8  FROM Customer C
  9  RIGHT JOIN Orders O
 10    ON C.Customer_ID = O.Customer_ID
 11  ORDER BY O.Order_Date;

ORDER_ID             ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                                        CUSTOMER_ID
-------------------- --------- ------------ ------------------------------------------------- --------------
CUSTOMER_NAME                               EMAIL
------------------------------------------- ------------------------------
ORD-505              24-MAR-26         1250 Completed                                         C004
Bob White                                   bob.white@email.com

ORD-506              25-MAR-26          720 Completed                                         C002
Jane Smith                                  jane.smith@email.com

ORD-507              26-MAR-26         1450 Baking                                            C003
Alice Brown                                 alice.brown@email.com

ORD-508              27-MAR-26          890 Pending                                           C001
John Doe                                    john.doe@email.com

ORD-509              28-MAR-26          950 Completed                                         C005
Charlie Green                               charlie.green@email.com
```

2. Find orders where customer details are missing.

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         O.Total_Amount,
  4         O.Order_Status
  5  FROM Customer C
  6  RIGHT JOIN Orders O
  7    ON C.Customer_ID = O.Customer_ID
  8  WHERE C.Customer_ID IS NULL;

no rows selected
```

3. Display all payment records with order details.

```
SQL> SELECT P.Payment_ID,
  2         P.Payment_Date,
  3         P.Payment_Mode,
  4         P.Payment_Status,
  5         P.Transaction_Amount,
  6         O.Order_ID,
  7         O.Order_Date,
  8         O.Total_Amount,
  9         O.Order_Status
 10  FROM Orders O
 11  RIGHT JOIN Payment P
 12    ON O.Order_ID = P.Order_ID
 13  ORDER BY P.Payment_Date;

PAYMENT_ID           PAYMENT_D PAYMENT_MODE
-------------------- --------- --------------------------------------------------
PAYMENT_STATUS                                     TRANSACTION_AMOUNT ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------------------------------------- ------------------ -------------------- --------- ------------
ORDER_STATUS
--------------------------------------------------
PAY-101              18-MAR-26 Credit Card
Completed                                                         660


PAY-102              19-MAR-26 UPI
Completed                                                         850


PAY-103              20-MAR-26 Debit Card
Completed                                                         360


PAY-104              22-MAR-26 Net Banking
Completed                                                         510


PAY-105              24-MAR-26 UPI
Completed                                                        1250 ORD-505             24-MAR-26         1250
Completed

PAY-106              25-MAR-26 COD
Completed                                                         720 ORD-506             25-MAR-26          720
Completed

PAY-108              26-MAR-26 Credit Card
Completed                                                        1450 ORD-507             26-MAR-26         1450
Baking

PAY-107              26-MAR-26 Debit Card
Failed                                                           1450 ORD-507             26-MAR-26         1450
Baking

PAY-109              27-MAR-26 Debit Card
Completed                                                         890 ORD-508             27-MAR-26          890
Pending


9 rows selected.
```

5. Retrieve Complete Order Details

Task

Generate a complete order report by combining:

- Customer Name
- Bakery Item Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         P.Title AS Bakery_Item_Name,
  3         OD.Quantity,
  4         O.Order_Date,
  5         O.Total_Amount,
  6         PM.Payment_Status
  7  FROM Customer C
  8  JOIN Orders O
  9    ON C.Customer_ID = O.Customer_ID
 10  JOIN Order_Details OD
 11    ON O.Order_ID = OD.Order_ID
 12  JOIN Product P
 13    ON OD.Product_ID = P.Product_ID
 14  LEFT JOIN Payment PM
 15    ON O.Order_ID = PM.Order_ID
 16  ORDER BY O.Order_Date, C.Customer_ID;

CUSTOMER_NAME
----------------------------------------
BAKERY_ITEM_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------------------
Bob White
Black Forest Cake
         2 24-MAR-26         1250 Completed

Jane Smith
Chocolate Fudge Cake
         1 25-MAR-26          720 Completed
```

6. Display Customer Purchase History

Tasks

1. All bakery items purchased by a customer.

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         O.Order_ID,
  3         O.Order_Date,
  4         P.Title AS Bakery_Item_Name,
  5         OD.Quantity,
  6         OD.Price
  7  FROM Customer C
  8  JOIN Orders O
  9    ON C.Customer_ID = O.Customer_ID
 10  JOIN Order_Details OD
 11    ON O.Order_ID = OD.Order_ID
 12  JOIN Product P
 13    ON OD.Product_ID = P.Product_ID
 14  WHERE C.Customer_ID = 'C001'
 15  ORDER BY O.Order_Date;

CUSTOMER_NAME                          ORDER_ID             ORDER_DAT
-------------------------------------- -------------------- ---------
BAKERY_ITEM_NAME
--------------------------------------------------------------------------------
  QUANTITY    PRICE
---------- --------
John Doe                               ORD-508              27-MAR-26
Chocolate Fudge Cake
         2   750.00
```

2. Total amount spent by each customer.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         ROUND(SUM(O.Total_Amount), 2) AS Total_Amount_Spent
  4  FROM Customer C
  5  JOIN Orders O
  6    ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  8  ORDER BY Total_Amount_Spent DESC;

CUSTOMER_ID          CUSTOMER_NAME                          TOTAL_AMOUNT_SPENT
-------------------- -------------------------------------- ------------------
C003                 Alice Brown                                          1450
C004                 Bob White                                            1250
C005                 Charlie Green                                         950
C001                 John Doe                                              890
C002                 Jane Smith                                            720
```

3. Number of orders placed by each customer.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         COUNT(O.Order_ID) AS Number_Of_Orders
  4  FROM Customer C
  5  LEFT JOIN Orders O
  6    ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  8  ORDER BY Number_Of_Orders DESC;

CUSTOMER_ID          CUSTOMER_NAME                                NUMBER_OF_ORDERS
-------------------- ---------------------------------------- ----------------
C002                 Jane Smith                                              1
C001                 John Doe                                                1
C004                 Bob White                                               1
C005                 Charlie Green                                           1
C003                 Alice Brown                                             1
C006                 David Miller                                            0

6 rows selected.
```

4. Latest purchase details of customers.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date AS Latest_Purchase_Date,
  5         O.Total_Amount,
  6         O.Order_Status
  7  FROM Customer C
  8  JOIN Orders O
  9    ON C.Customer_ID = O.Customer_ID
 10  WHERE O.Order_Date = (
 11      SELECT MAX(O2.Order_Date)
 12      FROM Orders O2
 13      WHERE O2.Customer_ID = C.Customer_ID
 14  )
 15  ORDER BY C.Customer_ID;

CUSTOMER_ID          CUSTOMER_NAME                            ORDER_ID             LATEST_PU TOTAL_AMOUNT
-------------------- ---------------------------------------- -------------------- --------- ------------
ORDER_STATUS
------------------------------------------------
C001                 John Doe                                 ORD-508              27-MAR-26          890
Pending

C002                 Jane Smith                               ORD-506              25-MAR-26          720
Completed

C003                 Alice Brown                              ORD-507              26-MAR-26         1450
Baking

C004                 Bob White                                ORD-505              24-MAR-26         1250
Completed

C005                 Charlie Green                            ORD-509              28-MAR-26          950
Completed
```

7. Generate Multi-Table Business Reports

Report 1: Customer Order Report

Display:

- Customer Name
- Order ID
- Order Date
- Order Status

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         O.Order_ID,
  3         O.Order_Date,
  4         O.Order_Status
  5  FROM Customer C
  6  JOIN Orders O
  7    ON C.Customer_ID = O.Customer_ID
  8  ORDER BY O.Order_Date;

CUSTOMER_NAME                          ORDER_ID             ORDER_DAT
-------------------------------------- -------------------- ---------
ORDER_STATUS
--------------------------------------------------
Bob White                              ORD-505              24-MAR-26
Completed

Jane Smith                             ORD-506              25-MAR-26
Completed

Alice Brown                            ORD-507              26-MAR-26
Baking

John Doe                               ORD-508              27-MAR-26
Pending

Charlie Green                          ORD-509              28-MAR-26
Completed
```

Report 2: Sales Report

Display:

- Bakery Item Name
- Quantity Sold
- Total Revenue

```
SQL> SELECT P.Title AS Bakery_Item_Name,
  2         SUM(OD.Quantity) AS Quantity_Sold,
  3         SUM(OD.Quantity * OD.Price) AS Total_Revenue
  4  FROM Product P
  5  JOIN Order_Details OD
  6    ON P.Product_ID = OD.Product_ID
  7  GROUP BY P.Product_ID, P.Title
  8  ORDER BY Total_Revenue DESC;

BAKERY_ITEM_NAME
--------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Chocolate Fudge Cake
            3          2250

Black Forest Cake
            2          1200
```

Report 3: Payment Analysis Report

Display:

- Payment Mode
- Number of Transactions
- Successful Payments

```
SQL> SELECT Payment_Mode,
  2         COUNT(*) AS Number_Of_Transactions,
  3         SUM(
  4             CASE
  5                 WHEN Payment_Status = 'Completed' THEN 1
  6                 ELSE 0
  7             END
  8         ) AS Successful_Payments
  9  FROM Payment
 10  GROUP BY Payment_Mode
 11  ORDER BY Number_Of_Transactions DESC;

PAYMENT_MODE                                  NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------------------------------------- ---------------------- ------------------
Debit Card                                                         3                  2
Credit Card                                                        2                  2
UPI                                                                2                  2
Net Banking                                                        1                  1
COD                                                                1                  1
```

Report 4: Customer Purchase Analysis

Top Purchasing Customers

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         SUM(O.Total_Amount) AS Total_Spending
  4  FROM Customer C
  5  JOIN Orders O
  6    ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  8  ORDER BY Total_Spending DESC;

CUSTOMER_ID          CUSTOMER_NAME                          TOTAL_SPENDING
-------------------- -------------------------------------- --------------
C003                 Alice Brown                                      1450
C004                 Bob White                                        1250
C005                 Charlie Green                                     950
C001                 John Doe                                          890
C002                 Jane Smith                                        720
```

Customers with Maximum Orders

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         COUNT(O.Order_ID) AS Number_Of_Orders
  4  FROM Customer C
  5  JOIN Orders O
  6    ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  8  HAVING COUNT(O.Order_ID) = (
  9      SELECT MAX(Order_Count)
 10      FROM (
 11          SELECT COUNT(Order_ID) AS Order_Count
 12          FROM Orders
 13          GROUP BY Customer_ID
 14      )
 15  );

CUSTOMER_ID          CUSTOMER_NAME                        NUMBER_OF_ORDERS
-------------------- ------------------------------------ ----------------
C002                 Jane Smith                                          1
C001                 John Doe                                            1
C004                 Bob White                                           1
C005                 Charlie Green                                       1
C003                 Alice Brown                                         1
```

Customers with Highest Spending

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  3         SUM(O.Total_Amount) AS Total_Spending
  4  FROM Customer C
  5  JOIN Orders O
  6    ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  8  HAVING SUM(O.Total_Amount) = (
  9      SELECT MAX(Total_Spending)
 10      FROM (
 11          SELECT SUM(Total_Amount) AS Total_Spending
 12          FROM Orders
 13          GROUP BY Customer_ID
 14      )
 15  );

CUSTOMER_ID          CUSTOMER_NAME                             TOTAL_SPENDING
-------------------- ---------------------------------------- --------------
C003                 Alice Brown                                         1450
```